

CONSOLIDATED ENGINEERING AND SOURCING CONSULTANTS (CESC)

Engineering White Paper

CT, VT, and CVT Selection for Grid Stations

Accuracy, Burden, Saturation, Ferroresonance, Transient Behaviour, and Scheme Suitability

Prepared, reviewed, and issued by CESC

Website publication edition

Executive abstract

This paper covers the application of instrument transformers in grid stations, including CT class selection, VT and CVT considerations, burden calculations, saturation limits, transient performance, and practical scheme suitability for measurement, protection, and metering. It also adds first-pass engineering checks that are commonly required during design review, relay coordination, FAT preparation, and commissioning.

Document basis

Document type	Technical white paper
Primary basis	IEC 61869-1, IEC 61869-2, IEC 61869-3, IEC 61869-5, IEC 60255 series
Intended audience	Utilities, EPC contractors, industrial asset owners, and engineering teams
Publication note	Prepared in formal issuer style to demonstrate CESC engineering capability in studies, protection, digital integration, retrofit, commissioning support, and asset management.

Applicable references

Reference	Application in this paper
IEC 61869-1	General instrument transformer requirements.
IEC 61869-2	Current-transformer application basis.
IEC 61869-3	Inductive voltage-transformer application basis.
IEC 61869-5	Capacitor voltage-transformer application basis.
IEC 60255 series	Protection relay functional context and application interface.

Purpose

Instrument transformer errors can compromise both protection dependability and metering confidence. Selection should therefore be driven by the connected scheme, burden, short-circuit duty, insulation level, and transient performance requirement rather than by nominal ratio alone.

This paper addresses CTs, inductive VTs, and capacitive voltage transformers in the context of grid-station design and explains where practical engineering checks should be carried out before procurement is frozen.

CT engineering themes

CT selection should consider continuous current, maximum through-fault current, relay and lead burden, class, short-time thermal duty, dynamic withstand, and where required knee-point performance. Protection cores, metering cores, and differential or busbar-dedicated cores should be treated separately.

For modern grid stations, it is poor practice to assume that one generic CT core can satisfy all functions. Metering, feeder overcurrent and earth fault, transformer differential, busbar differential, and disturbance recording may require different secondary performance assumptions.

A first-pass burden check is commonly expressed as $VA = I^2 \times R$ for the connected secondary circuit. The result helps verify whether the assumed burden and accuracy class remain realistic or whether conductor size, relay location, or CT class must be revised.

Practical CT example 1 - feeder protection burden check

Example basis: 400/1 A feeder CT, secondary leads 2 x 30 m of 2.5 mm² copper, relay input burden 0.15 ohm, terminal and contact allowance 0.05 ohm. Copper conductor resistance is approximated at about 7.4 ohm/km for 2.5 mm².

Lead loop resistance = 60 m x 7.4 ohm/km = 0.444 ohm. Total secondary resistance = 0.444 + 0.15 + 0.05 = 0.644 ohm.

At rated secondary current of 1 A, burden = $I^2 \times R = 1^2 \times 0.644 = 0.644$ VA. Even after adding design allowance, a 5 VA protection core is generally adequate on burden grounds. The engineering conclusion, however, must still check accuracy limit factor or knee-point requirement against the maximum through-fault current.

Practical value: this check often shows that the real burden is much lower than legacy assumptions, allowing more accurate class selection and reducing over-specification.

Practical CT example 2 - high-impedance busbar core check

Example basis: 2000/1 A dedicated busbar protection CT core used with a high-impedance scheme. Assume maximum external fault secondary current of 20 A, relay stabilizing resistor plus circuit resistance equivalent to 12 ohm, and a conservative safety factor.

Required knee-point voltage is often checked on a conservative first-pass basis as $V_k \geq 2 \times I_f \times (R_{ct} + R_{lead} + R_{relay})$. Using 20 A and 12 ohm total external circuit resistance gives a minimum indicative requirement of approximately 480 V before margin.

Engineering implication: if the selected CT knee point is too low, the scheme may become unstable for external faults. This is one reason busbar protection CTs should not be mixed casually with general-purpose feeder protection cores without a dedicated review.

VT and CVT application themes

VTs are generally preferred where transient accuracy and direct voltage reproduction are critical over a wide range, while CVTs remain common at higher transmission voltages where insulation economics, footprint reduction, and coupling functions support their use.

CVT application must consider transient response, ferroresonance risk, burden, frequency behavior, and the specific requirements of the connected protection functions, especially high-speed distance or line protection.

For synchronizing, metering, annunciation, and general SCADA voltage measurement, either VT or CVT application may be acceptable depending on system voltage and specification. For some fast protection applications, transient response must be reviewed more carefully.

Practical VT example - burden verification for metering and relays

Example basis: 11000/sqrt(3) V to 110/sqrt(3) V VT supplying one multifunction relay, one meter, one transducer, and wiring. Assume connected burdens of 8 VA + 5 VA + 3 VA + 4 VA = 20 VA total.

If the selected secondary winding is rated 100 VA, the operating burden is only 20 percent of rated burden. This is generally acceptable provided the accuracy class is specified for the intended metering function and the thermal burden remains within rating.

Practical value: the design package should identify each connected device by winding so that future additions do not unknowingly push one winding beyond its intended service.

Practical CVT example - scheme suitability check

Example basis: 132 kV line bay using CVTs for indication, metering, synchronizing, disturbance recording, and line distance protection. The initial engineering question is not only whether ratio and insulation are correct, but whether the transient response of the CVT is suitable for the exact relay application and reach philosophy.

Where permissive or high-speed distance schemes are used, the protection engineer should confirm relay manufacturer guidance for CVT transient compensation and review expected performance during faults close to the VT location. This is a scheme suitability exercise, not a simple catalogue selection exercise.

Selection framework by function

Metering cores should be selected for revenue or operational measurement accuracy at the relevant normal current range. Protection cores should be selected for stability and dependable operation under fault conditions. Differential or busbar cores require special attention to class, knee point, resistance matching, and saturation performance.

Similarly, VT and CVT windings should be allocated by function. A clear winding schedule should state which secondary serves protection, metering, synchronizing, power quality, or spare capacity. This prevents later site modifications from degrading critical functions.

Common engineering errors observed in projects

Common issues include selecting CT ratio only from normal load current, ignoring future feeder uprating, underestimating lead burden, allocating metering and protection to one unsuitable core, omitting knee-point checks for differential or busbar schemes, and specifying CVTs on fast protection schemes without a transient suitability review.

Other recurring errors include failing to reserve spare winding capacity, not documenting which devices are connected to which VT winding, and overlooking ferroresonance control or damping measures in the wider station design philosophy.

CESC engineering contribution

CESC can prepare instrument-transformer application schedules, burden calculations, class recommendations, core and winding allocation matrices, and scheme-specific suitability reviews that reduce late-stage problems during FAT and commissioning.

CESC value adding is strongest where a project needs more than catalogue selection: short-circuit study interface, relay philosophy alignment, busbar and differential CT review, VT/CVT suitability for line protection, burden optimization, future spare planning, and brownfield retrofit risk control.

In practical terms, CESC can translate system studies and protection philosophy into a defensible instrument-transformer specification, improving design clarity for vendors while reducing the chance of misapplication, nuisance operation, or commissioning delay.

Conclusion

These website papers are structured to demonstrate how CESC approaches engineering subjects at system level, not only at equipment catalogue level. The intended message to clients is that CESC can define studies, design criteria, scheme architecture, implementation strategy, testing approach, and brownfield execution method across protection, control, digital integration, retrofit, and asset management scopes.

For CT, VT, and CVT application in particular, the engineering value lies in selecting the correct device for the scheme, validating burden and transient behavior early, and documenting the purpose of every core and winding before procurement and commissioning. This is where disciplined CESC engineering adds measurable project value.